Adam B. Smith, Ph.D.

Center for Conservation & Sustainable Development,
Missouri Botanical Garden, 4344 Shaw Boulevard, Saint Louis, MO 63110 USA
314/577-9473 ext. 76314 | adam.smith@mobot.org | earthskysea.org | github.com/adamlilith

Education

Ph.D., Energy and Resources. Energy and Resources Group, University of California, Berkeley, CA. Advisor: John Harte. Dissertation title: "Facets of biological conservation: Appropriate science, policy, and ethics for ensuring the perpetuity of biodiversity."

M.A., Ecology. Curriculum in Ecology, University of North Carolina at Chapel Hill, Chapel Hill, NC. Advisor: Seth Reice. Thesis title: "Local versus regional processes in lotic macroinvertebrate communities: the control of species diversity across scales."

B.A., Biology (major), Math (minor). *Suma cum laude*. McPherson College, McPherson, KS.

Research Interests and Appointments

Research interests: My research addresses the impacts of global change on individual species and their communities and the means by which human intervention can facilitate adaptive responses. I use Bayesian models, artificial intelligence, spatial simulation, and analytical methods to address critical issues in conservation biogeography. A dominant theme of my work has been the development and application of models that predict the biogeographic responses of species to climate change, habitat loss, and expropriation by humans. Currently, my main interests lie in using Bayesian modeling to integrate multiple "facets" of species that we know are unified in nature—i.e., each species has a distribution, abundance, and sets of morphological and physiological traits—and yet are rarely combined analytically.

Scientist, Global Change Conservation, Center for Conservation and Sustainable Development, Missouri Botanical Garden, Saint Louis, MO. 2024 onward.

- Conducted research in issues in global change conservation by developing models to understand threats to biodiversity and prioritize conservation efforts at regional and global scales.

Associate Scientist, Global Change Conservation, Center for Conservation and Sustainable Development, Missouri Botanical Garden, Saint Louis, MO. 2020-2024.

Assistant Scientist, Global Change Conservation, Center for Conservation and Sustainable Development, Missouri Botanical Garden, Saint Louis, MO. 2014-2020.

Honorary Adjunct Faculty (concurrent appointment), Department of Biology, University of Washington in Saint Louis. 2023-onward.

Guest Affiliate Adjunct Professor (concurrent appointment), Department of Biology, University of Missouri—Saint Louis. 2026-onward.

Postdoctoral Fellow, Center for Conservation and Sustainable Development, Missouri Botanical Garden, Saint Louis, MO. 2011 to 2014.

- Conducted research evaluating the vulnerability of threatened plant species to climate change and climatic niche conservatism among species in the same clade. Supervisor: Matthew Albrecht.

Postdoctoral Scholar, Museum of Vertebrate Zoology, University of California, Berkeley, CA. 2009 to 2011.
- Validated species distribution models using historic vertebrate data collected by Joseph Grinnell. Assisted field teams in resurveys of historic mammal and bird census sites. Supervisors: Craig Moritz and Steve Beissinger.

Graduate Student Researcher, Energy & Resources Group and Institute of Governmental Studies, University of California, Berkeley, CA. 2005 to 2009.
- Collected field data, analyzed macroecological spatial datasets, compiled manuscripts.
- Compiled annotated bibliographies on biosafety and biodiversity governance, organized international workshops.

Research Assistant, Department of Environmental Science, Policy, and Management, University of California, Berkeley, CA. 2003 and 2005.
- Conducted research on global warming in subalpine meadows; surveyed native forbs; assayed soil chemistry.

Teaching Appointments

Adjunct Assistant Professor in Biology, University of Missouri at Saint Louis, Saint Louis, MO. Spring semester, 2015.
- Taught graduate-level Global Change Conservation course.

Graduate Student Instructor, Department of Integrative Biology and Energy & Resources Group, University of California, Berkeley, CA. 2003 to 2005.
- Taught Introductory Biology laboratory and discussion sections twice.
- Taught discussion sections of Quantitative Aspects of Global Environmental Problems.

Full-Time Volunteer, Trees for Life, Wichita, KS. 2002 to 2003.
- Developed ESL lessons for Global Circle of Knowledge peer-to-peer distance-based education software for villagers in India, Pakistan, and other underserved areas.

Assistant English Teacher, 風連町教育委員会 (Furen-cho Board of Education), Hokkaido Prefecture, Japan. 1998 to 2001.
- Taught English and international communications to kindergarten, elementary, junior high, and senior high students, and adults. Co-sponsored elementary and high school English clubs; participated in other clubs' activities (soft tennis, volleyball, tea ceremony, ping-pong, Japanese cooking, English). Led home stay with nine students in US for 2 weeks.

Assistant Professor of Biology, McPherson College, McPherson, KS. 1997 to 1998.
- Taught Senior Research, Research Methods, Conservation Biology, Environmental Science, Vertebrate Physiology, Evolution, Genetics, and College Biology I and II. Faculty sponsor for International Student Organization.

Graduate Teaching Assistant, University of North Carolina at Chapel Hill, Chapel Hill, NC. 1995 to 1996.
- Taught labs and recitations for Field Ecology once, and Ecology and Population Biology thrice.

Teaching Awards

Outstanding Graduate Student Instructor Award, University of California, Berkeley.

- Usually granted by course's host department only to its own graduate students, but exception made in granting it to me because I was in another department.

Professor of the Year, McPherson College.
- Youngest professor every to receive award.
- Only professor ever to receive award after just one year of service.

Tri-Beta Teaching Excellence Award in Biological Sciences, University of North Carolina at Chapel Hill. Nominated twice more for same.
- Granted to just one teaching assistant per year in a department with 3000 undergraduate majors.

Peer-Reviewed Publications

57) Pho*, J., Davis*, C., Kapp*, C., Guerrero*, E., Fogal, N.S., **Smith**, A.B., and Miller-Struttmann, N.E. 2026. Effects of habitat complexity on urban bee occupancy eclipse those of urbanization and flower diversity. *Biological Conservation* 321:111936. doi: 10.1016/j.biocon.2026.111936
* Undergraduate authors.
  - Interview by KMOX (starts at time 8:09 AM)
  - Story on St. Louis Public Radio.

56) Loza, M.I., Althaus, K., **Smith**, A.B., Cannon, C.H., Valencia-A., S., and Alvarez-Clare, S. 2026. Prioritization of areas for conservation of oaks (Fagaceae: *Quercus*) across their global biodiversity hotspot based on richness, endemism, and introgression. *Global Ecology and Conservation* 26:e04251. doi: 10.1016/j.gecco.2026.e04251

55) Condon, D., Levin, M.O., **Smith,** A.B., Morelli, T.L., Krasner, N.Z., Forester, E., Holmes, C., Narwold, B.P., Kalies, E.L., Wu, G.C., Gade, M.R., Kays, R., Robinson, F. and Hernandez, R.R. 2026. Practitioner perceptions of biodiversity criteria for solar suitability analyses in the United States. *npj Biodiversity* 5:24. doi: 10.21203/rs.3.rs-7437180/v1

54) **Smith**, A.B., Adams, E.G., Abercrombie, E., Bibi, N., Bottini, C.L.J., Brown, A.J., Cavalieri, C.N., Clauser, A.L., Duguid, M.C., Fowler-Finn, K., Hutchen, J., Jardim, V.L., Rodriguez, C.S., Swab, R.M. and Varghese, S. 2026. Ten simple rules for mentoring and being mentored while neurodiverse. *Public Library of Science Computational Biology* 22:e1013917. doi: 10.1371/journal.pcbi.1013917

53) **Smith**, A.B. 2026. *fasterRaster*: GIS in *R* using *GRASS* for large vectors and rasters. *Transactions in GIS* 30:e70238. doi: 10.1111/tgis.70238. Preprint doi: 10.31223/X52R0M.

52) Chang*, N.J., Eserman, L., Carmichael, A., **Smith**, A.B., Loy, X., Coffey, E.E.D., and Ojascastro, J. 2026. Ecological niche modeling reveals habitat differentiation and climatic vulnerability in two imperiled, sympatric southern Appalachian carnivorous plants. 2026. *American Journal of Botany* 113:e70194. doi: 10.1002/ajb2.70194. Preprint doi: 10.1101/2025.10.30.685572
* Undergraduate researcher.

51) Abercrombie, E., Myers, J., Usdin*, R.L., and **Smith**, A.B. 2026. Lagged responses in the composition of small mammal communities to a century of climate change. *Ecography* 2026:e08010. doi: 10.1002/ecog.08010 * Undergraduate author

50) Austin, M.W., Kaul, A.D., **Smith**, A.B., Rothendler, M., and Primack, R.B. 2026. Herbarium specimens reveal regional patterns of tallgrass prairie invasion and changing species abundance across 130 years. 2026. *New Phytologist* 251:881-895. doi: 10.1111/nph.70632
  - *State of the World's Plants and Fungi 2026* (Antonelli et al. 2026).
  - Part of virtual Joint Special Collection "Harnessing the benefits of specimen digitization" between *New Phytologist* and *Plants, People, Planet*.

49) Levin, M., Condon, D., Krasner, N.Z., Forester, E., Holmes, C.C., Bateman, B.L., Delach, A., Ennen, J.R., Kalies, E.L., Kays, R., Lovich, J.E., **Smith**, A.B., Wu, G.C., and Hernandez, R.R. 2025. Bibliographic synthesis of biodiversity-relevant criteria for solar energy siting. *Renewable & Sustainable Energy Reviews* 223:116026. doi: 10.1016/j.rser.2025.116026

48) Feng, X., **Smith**, A.B., Boyle, B., Chen, X., Enquist, B.J., Gallagher, R., Hammock, J., Ho, J.C., Lien, A.M., Maitner, B., Sokol, E.R., Soltis, P., Wenk, E.H., Willoughby, A., and Park, D.S. 2025. The next stage of biodiversity informatics: Community-driven synthesis and integration of biodiversity databases. *BioScience* 75:913-925. doi: 10.1093/biosci/biaf112

47) Beever, E.A., **Smith**, A.B., Wright, D. Rickman, T., Gerraty, F.A., Stewart, J.A.E., Gill, A., Klinger, K., and Robinson, M. 2025. Expanding barriers: Impassable gaps interior to distribution of an isolated mountain-dwelling species. 2025. *Ecosphere* 16:e70223. doi: 10.1002/ecs2.70223

46) Condon, D., Scott, T.A., **Smith**, A.B., Morelli, T.L., Ashraf, U., Mojica, A., Chittanuru, H., Luu, R., Bear, R., and Hernandez, R.R. 2025. Practitioners' perceived risks to biodiversity from renewable energy expansion through 2050. *Humanities and Social Sciences Communication* 12:263. doi: 10.1057/s41599-025-04558-9 and correction doi: 10.1057/s41599-025-04785-0

45) Beever, E.A., Westover, M., **Smith**, A.B., Gerraty, F., Billman, P., and Smith, F. 2025. Combining past and contemporary species occurrences with ordinal species distribution modeling to investigate responses to climate change. *Ecography* 2025:e07382. doi: 10.1111/ecog.07382

- Cover photo from the article in *Ecography*'s February 2025 issue.
- News coverage: Buckrail.

44) Kass, J.M., **Smith**, A.B., Warren, D.L., Vignali, S., Schmitt, S., Aiello-Lammens, M.E., de Andrade, A.F.A., Arlé, E., Barbosa, A.M., Broennimann, O., Cobos, M.E., Guéguen, M., Guisan, A., Merow, C., Naimi, B., Nenzén, H., Nobis, M.P., Ondo, I., Osorio-Olvera, L., Owens, H.L., Pinilla-Buitrago, G., Robertson, M., Sánchez-Tapia, A., Thuiller, W., Valavi R., Velazco, S.J.E., Zizka, A., Zurrell, D. 2025. Achieving higher standards in species distribution modeling by leveraging the diversity of available software. *Ecography* 2025:e07346. doi: 10.1111/ecog.07346

- Recognized as one of *Ecography's* top 10-most cited works published in in 2024.

43) Ashraf, U., **Smith**, A.B., Morelli, T.L., and Hernandez, R.R. 2024. Climate-smart siting for renewable energy expansion. *iScience* 27:110666. doi: 10.1016/j.isci.2024.110666 and correction doi: 10.1038/s41558-024-02216-7

42) Austin, M.W., **Smith**, A.B., Olsen, K.M., Hoch, P.C., Krakos, K.N., Schmocker, S.P., and Miller-Struttmann, N.E. 2024. Climate change increases flowering duration, driving phenological reassembly and elevated co-flowering richness. 2024. *New Phytologist* 243:2486-2500. doi: 10.1111/nph.19994

41) Naughtin*, S., Castilla, A.R., **Smith**, A.B., Strand, A.E., Dawson, A., Hoban, S., Abhainn, E.A., Romero-Severson, J., and Robinson, J.D. 2025. Integrating genomic data and simulations to evaluate alternative species distribution models and improve confidence in predictions of glacial refugia and future responses to climate change. *Ecography* 2025:e07196. doi: 10.1111/ecog.07196
* Undergraduate author

40) Ashraf, U., Morelli, T.L., **Smith**, A.B., and Hernandez, R.R. 2024. Aligning renewable energy expansion with climate-driven range shifts. *Nature Climate Change* 14:242-246. doi: 10.1038/s41558-024-01941-3

- Coverage by Spectrum News broadcast and online story.

39) Castilla A.R., A. Brown, S. Hoban, E.A. Abhainn, J.D. Robinson, J. Romero-Severson, A.B. **Smith**, A.E. Strand, J.R. Tipton, and A. Dawson. 2024 Integrative demographic modeling reduces

uncertainty in estimated rates of species' historical range shifts. *Journal of Biogeography* 51:325-336. doi: 10.1111/jbi.14754

38) Bartlett*, K., Austin, M.A., Beck, J., Zanne†, A., and **Smith**†, A.B. 2023. Beyond climate? Factors determining fruiting and flowering phenology across a genus over 117 years. *American Journal of Botany* 110:e16188. * Undergraduate author. † Co-senior authors. doi: 10.1002/ajb2.16188

- Subject of mini-documentary by HEC Media.
- Highlighted by the *American Journal of Botany*.
- Washington University in Saint Louis' press release.

37) Erickson, K.D. and **Smith**, A.B. 2023. Modeling the rarest of the rare: A comparison between multi-species distribution models, ensembles of small models, and single-species models at extremely low sample sizes. *Ecography* 2023:e06500. doi: 10.1111/ecog.06500

- Recognized as one of *Ecography's* top 10-most cited works published in in 2023.

36) Beever, E.A., Wilkening, J.L., Billman, P., Thurman, L., Ernest, K., Wright, D., Gill, A, Craighead, A., Helmstetter, N.A., Svancara, L.K., Camp, M.J., Bhattacharyya, S., Fitzgerald, J., Hirose, J.M.R., Westover, M., Gerraty, F., Klingler, K., Schmidt, D.A., Ryals, D.,. Brown, R., Clark, S., Clayton, N. Collins, G., Cutting, K., Doak, D., Epps, C.W., Foley, J., French, J. Hayes, C.L., Mills, Z., Moyer-Horner, L., Nichols, L., Orlofsky, K., Peacock, M., Penzel, N. Peterson, J., Ramsay, N., Rickman, T., Robinson, M., Robison, H. Rowe, K., Rowe, K., Russello, M., **Smith**, A.B., Stewart, J.E.A., Thompson, W., Thorne, J.H., Watterhouse, M., Weber, S., and Wilson, K. 2023. Geographic and taxonomic variation in adaptive capacity among mountain-dwelling small mammals: implications for conservation status and actions. *Biological Conservation* 282:109942. doi: 10.1016/j.biocon.2023.109942

35) **Smith**, A.B., Murphy, S.J., Henderson, D., and Erickson, K.D. 2023. Including imprecisely georeferenced specimens improves accuracy of species distribution models and estimates of niche breadth. *Global Ecology and Biogeography* 32:342-255. doi: 10.1111/geb.13628; open-access pre-print doi: 10.1101/2021.06.10.447988

- Recognized as one of *Global Ecology & Biogeography*'s top 10-most cited works published in in 2023.

34) Austin, M.A., Cole*, P., Olsen, K., and **Smith**, A.B. 2022. Climate change is associated with increased allocation to potential outcrossing in a common mixed mating species. *American Journal of Botany* 109:1085-1096. doi: 10.1002/ajb2.16021 * Undergraduate author.

- Highlighted by *The American Journal of Botany.*
- Covered by USA Today, *St. Louis Post Dispatch*, Missouri Botanical Garden's *Discover & Share*, St. Louis's FOX2, and Washington University in Saint Louis's Climate Change Program.

33) Johnson, L., Galliart, M., Alsdurf, J., Maricle, B.R., Baer, S.G., Bello, N.M., Gibson, D.J., and **Smith**, A.B. 2022. Reciprocal gardens as gold standard to detect local adaptation in grassland species: New opportunities moving into the 21st century. *Journal of Ecology* 110:1054-1071. doi: 10.1111/1365-2745.13695

- Recognized as one of *Journal of Ecology*'s Top Cited Articles 2022-2023.
- Included in the British Ecological Society's Key Concepts in Ecology reading list under the topic "Adaptations to variable environments".

32) Murphy, S.J. and **Smith**, A.B. 2021. What can community ecology learn from species distribution models? *Ecosphere* 12:e03864. doi: 10.1002/ecs2.3864

- One of *Ecosphere*'s most downloaded papers in the 12 months after publication.

31) Erickson, K.D. and **Smith**, A.B. 2021. Accounting for imperfect detection in data from museums and herbaria when modeling species distributions: Combining and contrasting data-level versus model-level bias correction. *Ecography* 44:1341-1352. doi: 10.1111/ecog.05679

30) Santos†, M.J., **Smith**†, A.B., Dekker, S.C., Eppinga, M.B., Leitão, P.J., Moreno-Mateo, D., Morueta-Holme, N., and Ruggieri, M. 2021. The role of land use and land cover change in climate change vulnerability assessments of biodiversity: A systematic review. *Landscape Ecology* 36:3367-3382. doi: 10.1007/s10980-021-01276-w † Equal contribution.

- Honorable Mention for *Landscape Ecology*'s Best Article of 2021.

29) Mohn, R.A., Oleas, N.H., **Smith**, A.B., Swift, J.F., Yatskievych, G., and Edwards, C. 2021. The phylogeographic history of a range disjunction in eastern North America: the role of post-glacial expansion into newly suitable habitat. *American Journal of Botany* 108:1042-1057. doi: 10.1002/ajb2.1686

- Highlighted as a noteworthy article (doi: 10.1002/ajb2.1690)

28) **Smith**, A.B., and Santos, M.J. 2020. Testing the ability of species distribution models to infer variable importance. *Ecography* 43:1801-1813. doi: 10.1111/ecog.05317

27) Morelli†, T.L., **Smith**†, A.B., Mancini, A.N., Balko, E.A., Borgerson, C., Dolch, R., Farris, Z., Federman, S., Golden, C.D., Holmes, S.M., Irwin, M., Jacobs, R.L., Johnson, S., King, T., Lehman, S.M., Louis Jr., E.E., Murphy, A., Randriahaingo, H.N.T., Randrianarimanana, H.L.Lucien, Ratsimbazafy, J., Razafindratsima, O.H., Baden, A.L. 2020. The fate of Madagascar's rainforest habitat. *Nature Climate Change* 10:89-96. doi: 10.1038/s41558-019-0647-x

† Equal contribution.

- Covered by *National Geographic*, *The New York Times*, *The Washington Post*, *The Conversation*, *ScienceDaily*, and EurekAlert!, amongst others.
- Altmetric: Top 1% of articles of the same age (as of December 2020).

26) Hoban, S., Dawson, A. Robinson, J., **Smith, A.B.**, Strand, A. 2019. Inference of biogeographic history by formally integrating distinct lines of evidence: genetic, environmental niche, and fossil. *Ecography* 42:1991-2011. doi: 10.1111/ecog.04327

- Runner-up article for *Ecography*'s annual E4 Award.

25) **Smith**, A.B., Beever, E.A., Kessler, A.E., Johnston, A.N., Ray, C., Epps, C.W., Lanier, H.C., Klinger, R.C., Rodhouse, T.J., Varner, J., Perrine, J., Seglund, A., Hall, E., Galbreath, K., Anderson, C., Billman, P., Blatz, G., Brewer, J., Vardaro, J.C., Chalfoun, A.D., Collins, G., Craighead, A., Curlis, C., Daly, C., Doak, D.F., East, M., Edwards, M., Erb, L., Ernest, K.A., Fauver, B., Foresman, K., Goehring, K., Hagar, J., Hayes, C.L., Henry, P., Hersey, K., Hilty, S.L., Jacobson, J., Jeffress, M.R., Manning, T.E., Masching, A., Maxwell, B., McCollough, R., McFarland, C., Miskow, E., Morelli, T,L., Moyer-Horner, L., Mueller, M., Nugent, M., Pratt, B., Rasmussen-Flores, M., Rickman, T.H., Robison, H., Rodriguez, A., Rowe, K.M.C., Rowe, K.C., Russello, M.A., Saab, V., Schmidt, A., Stewart, J.A.E., Stuart, J.N., Svancara, L.K., Thompson, W., Timmins, J., Treinish, G., Waterhouse, M.D., Westover, M.K., Wilkening, J., and Yandow, L. 2019. Alternatives to genetic affinity as a context for within-species response to climate. *Nature Climate Change* 9:787-794. doi: 10.1038/s41558-019-0584-8

- Cover story for *Nature Climate Change* Volume 9 Issue 10.
- Highlighted in editorial feature by Goldfather, M.F. 2019. Ecoregion shapes the range response. *Nature Climate Change* 9:730-731.
- Shortlisted for The Wildlife Society's Wildlife Publication Awards 2020.
- Covered by popular science websites including PBS NOVA, SciGlow, The Scientist, and PHYS.ORG.

24) **Smith**, A.B., Godsoe, W., Rodríguez-Sánchez, F., Wang, H-H., and Warren, D. 2019. Niche estimation above and below the species level. *Trends in Ecology and Evolution* 34:260-273. doi: 10.1016/j.tree.2018.10.012
    - Reviewed by Faculty of 1000.

23) Galliart, M., Bello, N., Knapp, M., Poland, J., St. Amand, P., Baer, S., Maricle, B., **Smith**, A.B., and Johnson, L.J. 2019. Local adaptation, genetic divergence, and experimental natural selection in a foundation grass across the US Great Plains' climate gradient. *Global Change Biology* 25:850-868. doi: 10.1111/gcb.14534

22) Frances, A., **Smith**, A.B., and Khoury, C. 2018. Conservation status and threat assessments for North American crop wild relatives. Pp. 189-208 in Greene S., Williams K., Khoury C., Kantar M., Marek L. (eds.) *North American Crop Wild Relatives: Conservation and Use, Volume 1*. (Springer, New York). doi: 10.1007/978-3-319-95101-0_7

21) Kunin, W.E., Harte, J., He, Fangliang, Hui, C., Jobe, R.T., Ostling, A., Polce, C., Šizling, A., **Smith**, A.B., Smith, Krister, Smart, S.M., Storch, D., Tjørve, E., Ugland, K-I., Ulrich, W., and Varma, V. 2018. Up-scaling biodiversity: Estimating the species-area relationship from small samples. *Ecological Monographs* 88:170-187. doi: 10.1002/ecm.1284

20) Santos, M.J., **Smith**, A.B., Thorne, J.H., and Moritz, C. 2017. The relative influence of change in habitat and climate on elevation range limits in small mammals in Yosemite National Park, California, U.S.A. *Climate Change Responses* 4:7. doi: 10.1186/s40665-017-0035-6

19) Kitzes, J., Berlow, E., Conlisk, E., Erb, K., Ihara, K., Martinez, N., Newman, E., Plutzar, C., **Smith**, A.B., and Harte, J. 2016. The global wildlife footprint: linking biodiversity loss to economic consumption. *Conservation Letters* 10:531-538. doi: 10.1111/con4.12321

18) **Smith**, A.B., Alsdurf, J., Knapp, M. and Johnson, L.C. 2017. Phenotypic distribution models corroborate species distribution models: A shift in the role and prevalence of a dominant prairie grass in response to climate change. *Global Change Biology* 23:4365-4375. doi: 10.1111/gcb.13666
    - Covered by at least 94 media outlets with a combined subscription/listener/watcher base of 78 million, including The San Francisco Chronicle, The Seattle Times, US News and World Report, The Topeka Capital Journal, The Manhattan Mercury, and numerous other regional newspapers, radio stations (e.g., KWMU 90.7), TV stations (e.g., KWCH12), and science and environmental news websites (e.g., Science News Online and NRDC)

17) Hernández-Yáñez, H., Kos, J.T., Bast, M.D., Griggs, J.L., Hage, P.A., Killian, A., Whitmore, M.B., Loza, M. L., **Smith**, A.B. 2016. A systematic assessment of threats affecting the rare plants of the United States. *Biological Conservation* 203:260-267. doi: 10.1016/j.biocon.2016.10.009
    - Covered by St. Louis Public Radio KWMU 90.7 (20 October 2016).
    - Featured on Inside Science (3 November 2016).

16) Ulrey, C., Quiantana-Ascencio, P.F., Kauffman, G., **Smith**, A.B., and Menges, E.S. 2016. Life at the top: Long-term demography, microclimatic refugia, and responses to climate change for a high-elevation southern Appalachian endemic plant. *Biological Conservation* 200:80-92. doi: 10.1016/j.biocon.2016.05.028

15) **Smith**, A.B., Long, Q.G., and Albrecht, M.A. 2016. Shifting targets: Spatial priorities for ex situ plant conservation depend on interactions between current threats, climate change, and uncertainty. *Biodiversity & Conservation* 25:905-922. doi: 10.1007/s10531-016-1097-7

14) Karger, D.N., Cord, A.F., Kessler, M., Kreft, H., Kühn, I., Pompe, S., Sandel, B., Cabral, J.S., **Smith**, A.B., Svenning, J-C., Tuomisto, H., Weigelt, P., and Wesche, K. 2016. Delineating probabilistic species pools in ecology and biogeography. *Global Ecology & Biogeography* 25:489-501. doi: 10.1111/geb.12422

13) **Smith**, A.B., M.J. Santos, M.S. Koo, K.C. Rowe, K.M.C. Rowe, J.L. Patton, S. Beissinger, and C. Moritz. 2013. Evaluation of species distribution models by resampling of sites surveyed a century ago by Joseph Grinnell. *Ecography* 36:1017-1031. doi: 10.1111/j.1600-0587.2013.00107.x

12) **Smith**, A.B. 2013. On evaluating species distribution models with random background sites in place of absences when test presences disproportionately sample suitable habitat. *Diversity & Distributions* 19:867-872. doi: 10.1111/ddi.12031

11) **Smith**, A.B. 2013. The relative influence of temperature, moisture, and their interaction on range limits of mammals over the past century. *Global Ecology & Biogeography* 22:334-343. doi: 10.1111/j.1466-8238.2012.00785.x

10) **Smith**, A.B., B. Sandel, N. Kraft, and S. Carey. 2013. Characterizing scale-dependent community assembly using the functional-diversity–area relationship. *Ecology* 94:2392-2402. doi: 10.1890/12-2109.1

9) Morelli, T.L., A.B. **Smith**, C. Kastely, I. Mastroserio, C. Moritz, and S. Beissinger. 2012. Anthropogenic refugia ameliorate the severe climate-correlated decline of a montane mammal along its trailing edge. *Proceedings of the Royal Society of London B* 279:4279-4286. doi: 10.1098/rspb.2012.1301
   - Covered by: *Science*, *Conservation Magazine*, *Science Daily*, *Grist*, and *Futurity*.

8) Barnosky, A.D, E.A. Hadly, J. Bascompte, E.L. Berlow, J.H. Brown, M. Fortelius, W.M. Getz, J. Harte, A. Hastings, P.A. Marquet, N.D. Martinez, A. Mooers, P. Roopnarine, G. Vermeij, J.W. Williams, R. Gillespie, J. Kitzes, C. Marshall, N. Matzke, D.P. Mindell, E. Revilla, A.B. **Smith**. 2012. Approaching a state-shift in Earth's biosphere. *Nature* 486:52-58. doi: 10.1038/nature11018
   - Altmetric: Top 1% of articles of the same age (as of December 2020).
   - Served as basis for *Scientific Consensus on Maintaining Humanity's Life Support Systems in the 21st Century*, a policy statement signed by >3400 scientists, presented to the US White House Staff, Secretary of State John Kerry, President Xi Jinping (China), Governor Ichiro Matsui (Osaka, Japan), Energy & Climate Change Minister Gregory Barker, (United Kingdom), Governor Eruviel Avila (Mexico), Right Honourable Datuk Seri Panglima Musa Haji Aman (Minister of Sabah, Malaysia), and others.
   - Served as basis of policy brief requested by Governor Jerry Brown (California). Served as supporting document for MOU on climate change between China's National Development and Reform Commission and California.
   - Covered widely in the national and international press, including *The New York Times*, *BBC World News*, *Los Angeles Times*, WIRED, and Bill Moyers; critiqued in *Al Jazeera* op-ed.
   - Subject of documentary *Demain* (released as *Tomorrow* in English) by Mélanie Laurent and Cyril Dion, shown in public theaters and UNFCCC COP21, UNESCO, and other diplomatic venues.
   - Reviewed by Faculty of 1000.
   - Used as basis for Consortium for Mathematics' 2013 Contest in Modeling.

7) **Smith**, A.B. 2010. Caution with curves: Caveats for using the species-area relationship in conservation. *Biological Conservation* 143:555-564. doi: 10.1016/j.biocon.2009.11.003

6) Harte, J., A.B. **Smith**, and D. Storch. 2009. Biodiversity scales from plots to biomes with a universal species-area curve. *Ecology Letters* 12:798-797. doi: 10.1111/j.1461-0248.2009.01328.x

5) Sandel, B. and A.B. **Smith**. 2009. Scale as a lurking factor: Incorporating scale-dependence in experimental ecology. *Oikos* 118:1284-1291. doi: 10.1111/j.1600-0706.2009.17421.x

4) Harte, J., Z.T. Zillo, E. Conlisk, and A.B. **Smith**. 2008. Maximum entropy and the state variable approach to macroecology. *Ecology* 89:2700-2711. doi: 10.1890/07-1369.1

- Reviewed by Faculty of 1000 (two recommendations).

3) Srinivasan, U.T., S.P. Carey, E. Hallstein, P.A.T. Higgins, A.C. Kerr, L.E. Koteen, A.B. **Smith**, R. Watson, J. Harte, and R.B. Norgaard. 2008. The debt of nations and the distribution of ecological impacts from human activities. *Proceedings of the National Academy of Sciences USA* 105:1768-1773. doi: 10.1073/pnas.0709562104
   - Covered by *Nature*, *The Guardian*, and *Monga Bay*.

2) Harte, J., E. Conlisk, A. Ostling, J.L. Green, A.B. **Smith**. 2005. A theory of spatial structure in ecological communities at multiple spatial scales. *Ecological Monographs* 75:179-197. doi: 10.1890/04-1388

1) Lloyd-Smith, J.O., P.C. Cross, C.J. Briggs, M. Daugherty, W.M. Getz, J. Latto, M.S. Sanchez, A.B. **Smith**, and A. Swei. 2005. Should we expect population thresholds for wildlife disease? *Trends in Ecology & Evolution* 20:511-519. doi: 10.1016/j.tree.2005.07.004

Submitted/in review/in revision:

- Thonis, A., Cavalieri, C.N., Morelli, T.L., Ashraf, U., Lee, A.K.C., Jiménez-Valverde, A., Zarzo-Arias, A., Ascanio, A., Kramer, A.M., Smith, A.M., Naimi, B., Rose, M.B., Ramirez-Reyes, C., Milburn, C., Zurell, D., Warren, D., Nemer, D., Wilkinson, D.P., Arlé, E., Lim, F.K.S., Owens, H.L., Ondo, I., Dahan, I., Kass, J.M., Franklin, J., Bracken, J., Svenning, J.-C., Belmaker, J., Serra-Diaz, J.M., White, J.D.M., Zhu, K., Farleigh, K., Ash, L.V., Aguirre-Lopez, L.J., Pertierra, L.R., Ledger, M.J., Fitzpatrick, M.C., Nobis, M.P., Zimmermann, N.E., Hepner Ucko, O., Peña-Aguilera, P.O., Title, P., Aragón, P., Jinga, P., Brun, P., Valavi, R.M., Chefaoui, R.M., Mateo, R.G., Pienaar, R., Pironon, S., Velazco, S.J.E., Volkel, S., Leão, T., Jezkova, T., Bonebrake, T.C., Feng, X., Chen, X., Song, Y., **Smith**, A.B. Is species distribution modeling converging as a field? Consensus and divergence among teams in a blinded experiment.
- Richardson*, N., Vilna-Santos*, L., Cortes**, A., Gaughan, M.K., **Smith**, A.B., and Austin, M.W. Phenological and stomatal responses of herbaceous plants to climate change since the 1800s: A case study with herbarium specimens. * Undergraduate author. ** High school author. pre-print doi: 10.32942/X2MM6P
- Ashraf, U., **Smith**, A.B., Hernandez, R.R., Bateman, B.L., Gedir, J.V, Grand, J., and Morelli, T.L. Climate-smart wind energy development for North American bird conservation.
- Sytsma, J., Johnson, L., Louthan, A., Newman, E., **Smith**, A.B., Winters, H., Galliart, M., Lee, S., Jumpponen, A., and Vogt, B. Testing core vs. margin paradigms: Abundance, traits, and genetic patterns of a dominant grass across its broad range.
- Beever, E.A., Westover, M.L., **Smith**, A.B., Gerraty, F.D., Billman, P.D., Smith, F.A. Investigating patterns of density and distributional decline in a habitat specialist not experiencing physical-habitat loss, to inform conservation decisions.
- Austin, M., Panchen, Z., Corlett, R., Ahrendts, A., Brant, R., Fu, Y., Gröger, A., Hamid, M., Ahlstrand, N.I., Jung, M., Kim, H., Kleissenberg, N., Lautenschläger, T., Lee, S., Magee, G., McCormack, M.L., Nikitin, T., Nordt, B., Parolly, G., Rauschkolb, R., Reidy, B., Riehn, J., Rollinson, C., Römermann, C., Rosche, C., Shafee, F., Shah, M., Shearer, K., **Smith**, A.B., Son, S., Träger, S., Umazekabiri, R., Wirth, C., Primack, R. Leaf marcescence in woody plants is influenced by phylogeny, plant size, age, vertical stratification, and climate.
- Westover, M.L., Beever, E.A., **Smith**, A.B., Gerraty, F.D., Billman, P.D., and Smith, F.A. Climate outweighs biogeography in governing density and multi-scale distributional patterns across a trailing-edge mainland.

Peer-reviewed Reports

Hart, M., Johnson, M., Mitchell, J., Rock, R., Shafron, E., Tobalske, C., Banad, S., Blair, M., Borokini, I., Burgos, N., Burns, P., Chardon, N., Copenhaver-Parry, P., Easterday, K., Feng, X., Fitzpatrick, M., Gerstner, B., Goljani-Amirkhiz, R., Hakkenberg, C., Howard, T., Jantz, P., Kumar, A., Liu, Z., Martin, E. C., Merow, C., Nielsen, E., Rapacciuolo, G., Roy, S., Rush, S., Schwizer, L., **Smith**, A. B., Wan, H. Y., Zhai, L. 2022. *Recommendation Report: Broadening the Use of NASA Datasets by the Species Distribution Modeling (SDM) Community*. NASA Understanding User Needs to Broaden Outside Use of NASA Data (UNBOUND) Program. doi: 10.13140/RG.2.2.16819.60960

Huntley, B., Foden, W.B., **Smith**, A.B., Platts, P., Watson, J. and Garcia, R.A. 2016. Chapter 5. Using CCVAs and interpreting their results. In W.B. Foden and B.E. Young, editors. *IUCN SSC Guidelines for Assessing Species' Vulnerability to Climate Change*. Version 1.0. Occasional Paper of the IUCN Species Survival Commission No. 59. Gland, Switzerland and Cambridge, UK. pp 33–48. doi: 10.2305/IUCN.CH.2016.SSC-OP.59.en

Huntley, B., Foden, W.B., Pearce-Higgins, J., and **Smith**, A.B. 2016. Chapter 6. Understanding and working with uncertainty. In W.B. Foden and B.E. Young, editors. *IUCN SSC Guidelines for Assessing Species' Vulnerability to Climate Change*. Version 1.0. Occasional Paper of the IUCN Species Survival Commission No. 59. Gland, Switzerland and Cambridge, UK. pp 49–56. doi: 10.2305/IUCN.CH.2016.SSC-OP.59.en

Brandt, L., He, H., Iverson, L., Thompson, F., Butler, Patricia, Handler, S., Janowiak, M., Swanston, C., Albrecht, M., Blume-Weaver, R., Dijak, B., Deizman, P., DePuy, J., Dinkel, G., Fei, S., Jones-Farrand, T., Leahy, M., Matthews, S., Nelson, P., Oberle, B., Perez, J., Peters, M., Prasad, A., Schneiderman, J.E., Shuey, J., **Smith**, A.B., Studyvin, C., Tirpak, J., Walk, J., Wang, W., Watts, L., Weigel, D., Westin, S. 2014. Central Hardwoods ecosystem vulnerability assessment and synthesis: A report from the Central Hardwoods Climate Change Response Framework project. General Technical Report NRS-124, Newtown Square, PSA, U.S. Department of Agriculture, Forest Service, Northern Research Station. doi: 10.2737/NRS-GTR-124

Commissioned and Invited Works and Reports, and Contributions Thereto

Antonelli, A., Fry, C., Smith, R.J., Eden, J., Kersey, P.J., Larridon, I., Paton, A.J., Simmonds, M.S., Ainsworth, A.M., Bachman, S., Cheek, M., Davis, C., Forest, F., Lennon S., Little, D.P., Nesbitt, M., Nic Lughadha, E., Smith, S., Wilkinson, T., Williams, et al. (261 authors). 2026. *State of the World's Plants and Fungi 2026*. Royal Botanic Gardens, Kew. doi: 10.34885/rdzs-ky83

Primack, R.B., Austin, M.W., Corlett, R.T., Panchen, Z.A., Gröger, A., Fu, Y.H., Nordt, B., Parolly, G., Lee, S., Rauschkolb, R., Römermann, C., Träger, S., Umazekabiri, R., Wirth, C., Kim, H., Son, S., Rosche, C., Brant, R.A., Riehn, J.K., **Smith**, A.B., McCormack, M.L., Nikitin, T., Reidy, B.C., Rollinson, C.R., Shearer, K., Iwanycki Ahlstrand, N., Jung, M., Kleissenberg, N., Lautenschläger, T., Shah, M.A. 2025. Informal research networks of botanical gardens. *BGjournal* 22:10-14.

Smith, A.B. 2023. Where will Midwest ecosystems be in 2500? *Missouri Naturalist Newsletter* 23:22-27.

Smith, A.B. 2020. Species-area relationship. *30-Second Ecology*. Ivy Press.

Smith, A.B. 2020. Island biogeography. *30-Second Ecology*. Ivy Press.

Smith, A.B., Albrecht, M.A., and Hird, A. 2014. "Chaperoned" managed relocation. *BGjournal* 11:19-22.

Rowe, K.M.C., C. Moritz, S.J. Beissinger, A.B. **Smith**, and M.S. Koo. 2012. Region-wide planning for resurvey of vertebrate diversity over the last 100 years in the Great Basin and Mojave Desert. 2012 Final Report. Natural Resource Technical Report NPS. National Park Service, Fort Collins, Colorado.

Smith, A.B., M. Koo, K.C. Rowe, J., Patton, S. Beissinger, and C. Moritz. 2011. Testing methods for predicting mammalian species' responses to 20th century climate change in California. California Energy Commission, Public Interest Energy Research Program.

Smith, A.B. 2011. Convention on Migratory Species of Wild Animals. *Encyclopedia of Environmental Issues*, Salem Press, Pasadena.

Smith, A.B. 2011. International Convention for the Regulation of Whaling. *Encyclopedia of Environmental Issues*, Salem Press, Pasadena.

Smith, A.B. 2011. Sea Shepherd Conservation Society. *Encyclopedia of Environmental Issues*, Salem Press, Pasadena.

Smith, A.B. 2009. International biodiversity governance and the outpacing of policy by threats: How can conservation regimes address global climate change? For "Managing Biosafety and Biodiversity in a Global World: EU, US, California and Comparative Perspectives," Center for Institutions and Governance, UC Berkeley and Centre for Global Governance Studies, Katholieke Universiteit Leuven (€3750).

Smith, A.B. 2009. International Union for the Conservation of Nature. In Ducth, S.I (ed.) *Encyclopedia of Global Warming*, Salem Press.

Smith, A.B. 2009. Dolphins and porpoises. In Ducth, S.I (ed.) *Encyclopedia of Global Warming*, Salem Press.

Grants, Contracts, Fellowships, and Scholarships

Total: >$4,000,000 leveraging >$4,450,000 of assets.

"Biodiversity on Demand (BOND): A web platform for discovering, comparing, and integrating biodiversity databases," National Science Foundation, 2026-2029, $187,990.

"Eco-evolutionary forces responsible for patterns of micro-endemism in an island biodiversity hotspot," National Science Foundation, 2025-2028, $377,986.

Phase II NSF Convergence Accelerator Funding Working Group: "AI for Coral Reef Observation and Forecasting, Resilience, and Adaptation", 2023.

Phase I NSF Convergence Accelerator Training program: "AI for Climate Change", 2023.

"Translational Climate-Smart Siting for Renewable Energy: Anticipating Future Conservation Risks and Opportunities for Vulnerable Species," Sloan Foundation, 2022-2025, $490,000.

"Comparison of methods for estimating past glacial refugia and subsequent migration by trees: Pollen-vegetation models versus ecological niche modeling," NSF, 2021-2022, supplement to NSF ABI, post-bachelorette award, $46,076.

"Conservation Stewardship Legacy for Porcupines and Chipmunks," USGS, 2021-2023, $69,000.

"Genetic and phenotypic variation of a dominant forage grass across the Great Plains and modeling predicted response to climate," USDA NIFA, 2021-2023, $500,000 (with Loretta Johnson, Erliang Zeng, Brad Olson, and Matthew Galliart).

"Linking microbiome function and microbial processes to plant genetic diversity in a foundation forage grass across the Great Plains grassland climate gradient: A multi-omics approach," USDA NIFA, 2020-2023, $750,000 (with Sonny Lee, Sara Baer, Loretta Johnson, Ari Jumpponen).

"Using species distribution models to predict local-scale habitat suitability and species occurrence," Living Earth Collaborative of Washington University in Saint Louis, 2019-2020, $28,740 (co-PI with Stephen Murphy, Kim Medley, and Jonathan Myers).

"Collaborative Research: ABI Innovation: Quantifying biogeographic history: a novel model-based approach to integrating data from genes, fossils, specimens, and environments," National

Science Foundation, Advances in Biological Informatics, 2018-2021, $698,757 (co-PI with Sean Hoban, Allan Strand, Andria Dawson, and John Robinson).

"Inferring the Invisible: Extending the Use of Natural History Museum and Herbarium Specimens Beyond the Locations in which They Were Collected," Institute for Museum and Library Services, 2015-2020, $351,579 (with Iván Jiménez).

"Developing General Methods of Inferring Vulnerability to Climate Change Using Well-Known Iconic Species in the American West", US Geological Survey, 2015-2016, $23,500.

Graduate Travel Grant ($500), Summer 2007, University of California, Berkeley Graduate Fellowships Office, Berkeley, California.

Mildred E. Mathias Graduate Student Research Grant ($1500), Spring 2007. University of California Natural Reserve System, Oakland, CA.

Mildred E. Mathias Graduate Student Research Grant ($450), Spring 2006. University of California Natural Reserve System, Oakland, CA.

Energy and Resources Block Grant (full tuition and stipend, 1 semester), Fall 2004. Energy and Resources Group, University of California, Berkeley.

Datatel Returning Student Scholarship ($1,500), 2003. Datatel Scholars Foundation, Datatel, Fairfax, VA.

Graduate Student Fellowship ($8500, 1 year), 1994-1995, The Graduate School, University of North Carolina at Chapel Hill, Chapel Hill, NC.

Honorary Full-time Scholarship (full tuition, 4 years), McPherson College, McPherson, KS.

Invited Presentations (selected)

"How to Protect Earth's Biological Diversity Forever: Lessons from Maitreya Buddha, Pharma Bros, and Yucca Mountain." Saint Louis Academy of Science, 2020 & Bethel College, 2026.

"Overcoming the Hutchinsonian shortfall: Improving our understanding of species' fundamental niches." Oklahoma State University, 2024.

"Reconstructing species' biogeographic histories by integrating species distribution models, pollen density models, and demographic/genetic simulations." Modeling in Ecology and Evolution Meeting, 2022.

"Addressing the Hutchinsonian shortfall: Improving our understanding of species' fundamental environmental tolerances." Washington University in Saint Louis, 2022.

"Endangered species: Cataloguing the risks to life." National Federation of Garden Clubs, 2022.

"Intraspecific variation in species' responses to climate change: Phenotype, phylogeny, and place." Washington University in Saint Louis, 2017.

"Rearranging deck chairs on the Titanic amidst low-hanging fruit: A systematic assessment of threats to rare plants of the United States." Wichita State University, 2016.

"Climate change and biodiversity: Out or up!" Confluence Chapter of the Missouri Master Naturalists, 2016.

"Death by 1000 cuts while being loved to death amidst low-hanging fruit: Threats facing the rare plants of the United States." University of Missouri at Saint Louis, 2015.

"Global change, climate change, and vulnerability of rare and threatened plants of the North American Interior Highlands." Kansas State University, 2014.

"Ecological fortunetelling: Validating Predictions of Ecological Change Using Historic Data Collected by Joseph Grinnell." Saint Louis University, 2013.

"Using GIS to Predict the Vulnerability of Plants to Climate Change." Missouri GIS Conference, 2013.

"Ecological Fortunetelling: Projecting Species Distribution Models across Time Spans Relevant to Conservation." Kansas State University, 2012.

"Over Par: Fairways and Sand Traps on the Migration Route to a Favorable Climate." Saint Louis Consortium for Ecology, Evolution, and Conservation, 2012.

"Biogeographic Vulnerability of Rare Plants of the Central Highlands and Central Basin." US Forest Service Central Hardwoods Climate Change Vulnerability Assessment Workshop, 2012. (With Matthew A. Albrecht.)

"California Climate" and "Species' Responses to Climate Change: Adaptation, Range Shifts, and Extinction" for the Sierra Nevada Climate Change Institute (California Institute of Biodiversity), June 2010 ($1500).

"Human Impacts on the Global Environment" (three lectures) for Quantitative Aspects of Global Environmental Problems for Policymakers, Monterey Institute of International Studies, Monterey, California, 2010 ($900).

"Ecological Debt and the Debt of Whaling" for Environmental Science, California College of the Arts, Oakland, California, 2009 and 2010 ($150 each).

"$\lim_{a\to\infty} H \int_r^a \sinh \sqrt[te]{\text{☺}d\text{☺}}$." Three Bridges: A Symposium in Honor of John Harte, Rocky Mountain Biological Laboratory, 2009.

"International biodiversity governance and complex problems: How can biodiversity conservation regimes address global warming?" University of Leuven, Belgium. Managing Biosafety and Biodiversity in a Global World – EU, US, California, and Comparative Perspectives, 2009.

Conference Abstracts (listing only presentations where I was the main presenter)

"Variation across species distribution models of the same species modeled by independent teams: Are we converging as a field?" International Biogeography Society Meeting, Åarhus, Denmark, 2026.

"What is the useful 'lifetime' of a species distribution model used in conservation decision-making? Keeping pace with rapidly changing methods and standards," Ecological Society of America, Portland, Oregon, 2023.

"Geographic niche truncation limits our ability to foresee the future with species distribution models… so let's look to the past," Species on the Move, Bonita Springs, Florida, 2023.

"Standards for choosing the appropriate background for calibrating species distribution models and ecological niche models," INTECOL, Geneva, 2022.

"Recreating species' biogeographic histories over the last 21,000 years using an integrated species distribution/pollen-vegetation model," Annual meeting of the Ecological Society of America in 2022 (with Andria Dawson, Sean Hoban, John Robinson, Antonio Castilla, Allan Strand, Alissa Brown, and Lauren Jenkins).

"Natural niche truncation hinders accurate prediction of species' past and future responses to changing climates," Annual meeting of the Ecological Society of America in 2021 (with David Henderson).

"Integration of data types for inferring species' biogeographic histories: Genetic, fossil, and occurrence," Annual meeting of the Ecological Society of America in 2020 (with Andria Dawson, Alissa Brown, Allan Strand, Sean Hoban, and John Robinson).

"Five billion six hundred seventy million years from now: Perspectives on extinction," Crafting the Long Tomorrow symposium in 2019, Biosphere 2, Tucson, Arizona.

"Extending the 'species distribution model' paradigm to ecological phenotypes", Annual meeting of the Ecological Society of America 2017, Portland, Oregon (with Loretta Johnson, Jacob Alsdurf, Mark Knapp, and Sara Baer).

"Non-stationary control over the range limits of species inferred from spatially-distributed ecological niche models." International Biogeography Society, Tucson, 2017 (with Erik Beever, Mimi Kessler, Aaron Johnston, and 72 others).

"Efficient conservation seed banking in light of current threats, threats from future climate change, and uncertainty." Global Partnership for Plant Conservation, Saint Louis, Missouri, 2016.

"What can global whaling negotiations tell us about global warming negotiations?" Saint Louis Ecology, Evolution, and Conservation Symposium, 2015.

"Using species distribution models to infer the importance of factors limiting species' ranges." Annual meeting of the Ecological Society of America 2015, Baltimore, Maryland (with Maria J. Santos).

"Botanical gardens as networks for 'chaperoned' assisted migration of the world's flora." Annual meeting of the Ecological Society of America 2013, Portland, Oregon (with Matthew A. Albrecht).

- Covered by *Nature*: Gewin, V. 2013. Plan seeks 'chaperones' for threatened species: Botanical gardens proposed as stopping-off points for plant species as climate warms.
- Featured in *Scitable* (*Nature Education*): Wittington, K. 2013. Plant pioneers: Assisting the migration of climate-endangered species.

"Prediction of Migratory Networks from Climate Change Velocity." International Biogeography Society meeting, 2013.

"Multivariate Climate Change Velocity of Ecoregions and Threatened Species." The First North American Congress for Conservation Biology (with Matthew A. Albrecht and Quinn G. Long.)

"Validating species distribution models with data collected a century ago by Joseph Grinnell." Biennial meeting of the San Francisco Bay Area Society for Conservation Biology, 2011.

"Predicting the ranges of species across time spans relevant to conservation: Species distribution modeling of vertebrates observed by Joseph Grinnell and contemporary resurveys." Annual meeting of the Ecological Society of America, 2010 (with C. Moritz et al.)

"Comparing explanations for the species-area relationship: What determines diversity across multiple spatial scales?" Annual meeting of the Ecological Society of America, 2008.

"Developing a mechanistic understanding of the species-area relationship from serpentine habitats." Biennial Mildred E. Mathias Symposium, 2008.

"Comparing theory to data: What, when, and how we explain the species-area relationship." Annual meeting of the Ecological Society of America, 2007 (with John Harte).

Academic and Professional Service Affiliations

Biodiversity Fellow, Living Earth Collaborative, Washington University in Saint Louis, since 2018.

Member, Ecological Society of America, since 1996.

Member, Society for Conservation Biology, since 2003.

Member, Climate Change Specialist Group, Species Survival Commission, International Union for the Conservation of Nature, since 2016.

Affiliate, Whitney R. Harris World Ecology Center, since 2015.

Associate Editor, *Biological Invasions*, 2016-2019.

Computational Proficiencies

Programming languages: R (advanced). Developer and maintainer of four R packages on CRAN:

- `fasterRaster`: Faster Raster and Spatial Vector Processing Using GRASS GIS
- `enmSdmX`: Species Distribution Modeling and Ecological Niche Modeling
- `statisfactory`: Statistical and Geometrical Tools
- `omnibus`: Helper Tools for Managing Data, Dates, Missing Values, and Text

Geographical analysis: R, GRASS, QGIS, and ArcGIS.

Workshops/Seminars/Webinars Led

"Biosphere 2 Biogeography," an NSF-sponsored workshop on reconstructing historical biogeography through data integration (with Sean Hoban, Andria Dawson, Allan Strand, John Robinson, Antonio Castilla; 2023).

"Neurodiversity in ecology for mentors and mentees: Intraspecific diversity in facets of thinking," a special session at the annual meeting of the Ecological Society of America (with Emily Morris; 2022).

"Modeling species distributions and niches," a 7-hour video curriculum for the Center for Plant Conservation's course on plant conservation (2021).

"Introduction to retroactive georeferencing of herbarium and museum specimens" webinar (October 2021).

"Using Museum and Herbarium Specimen Data in Ecology, Evolution, and Conservation"

- A free webinar attended by >400 full-length viewers worldwide (April 2020).

"Species Distribution Modeling in R: From Start to Finish"

- Kansas State University, 2012 and 2016
- Wichita State University, 2016
- North American Congress for Conservation Biology, 2012 and 2016
- Ecological Society of America, 2017
- Missouri Botanical Garden, 2012

"Methods for inferring biogeographic history from distinct lines of evidence: genetic, environmental niche, and fossil" workshop at the International Biogeography Society, Quito, Ecuador, 2019 (with Sean Hoban, Allan Strand, Andria Dawson, John Robinson, and Danielle Jensen-Ryan)

Mentoring

Postdoctoral scholars:

Uzma Ashraf, University of California, Davis: "Translational climate-smart siting for renewable energy: Anticipating future conservation risks and opportunities for vulnerable species." 2021-ongoing.

Isabelle "Bella" Loza, Morton Arboretum: "Identifying priority areas for conservation of oaks in their evolutionary ancestral range." 2023-2025.

Matthew Austin, Washington University in Saint Louis: "Eco-evolutionary effects of phenological shifts: Pollen transfer and evolution of self-fertilization." 2020-ongoing.

Kelley Erickson, Missouri Botanical Garden: "Inferring the invisible: Extending the use of natural history museum and herbarium specimens beyond the locations and times they were collected." 2018-ongoing.

Stephen Murphy, Missouri Botanical Garden: "Inferring the invisible: Extending the use of natural history museum and herbarium specimens beyond the locations and times they were collected." 2018-2020.

Camilo Sanín, Missouri Botanical Garden: Developing next-generation species distribution models for erstwhile unusable data using Bayesian methods. 2016-2018.

Post-graduates

Haydée Hernández-Yáñez, University of Missouri—Saint Louis: Assessing threats to the rare plants of the United States. Post-master's advisor, 2014-2015.

Vettes Kalema, post-graduate AWARD Fellow, University of Kampala, Uganda: Assessing climate change vulnerability of woodlands used for indigenous charcoal production in Nakasongola District, Uganda. Scholarship mentor, 2011-2012.

Ph.D. students

Mia Gaughan, Washington University in Saint Louis, Ph.D. student: Primary mentor, 2025 onward.

Jillian Cieslik, Washington University in Saint Louis, Ph.D. student: Primary co-mentor and lab rotation mentor. 2025 onward.

Ezra Tasman Rosenfeld, Washington University in Saint Louis, Ph.D. student: Lab rotation mentor. 2025.

Daphne Concon, University of California, Davis, Ph.D. student: Ph.D. committee member. 2025-onward.

Arielle Xiaofen Lu, University of Arizona, Ph.D. student: Dissertation chapter mentor. 2025-present.

Sean McHugh, Washington University in Saint Louis, Ph.D. student: Ph.D. committee member. 2023-present.

Ethan Abercrombie, Washington University in Saint Louis, Ph.D. student: Lab rotation mentor and Ph.D. committee chair. 2021-present.

Michael Ngoh, University of Tennessee—Knoxville, Ph.D. student: Dissertation chapter mentor. 2022-2025.

Quinn Fox, Washington University in Saint Louis: Ph.D. committee member. 2021-2023.

James Lucas Ojascastro, Washington University in Saint Louis: Modeling present and future potential ranges of plants used for papermaking. Ph.D. committee member. 2020-2023.

David Henderson, Washington University in Saint Louis: Lab rotation mentor and Ph.D. committee member. 2019-2022.

Holly Bernardo, Washington University in Saint Louis: Computational tools for conservation of rare plants. Ph.D. committee member.

Jennifer Gruhn, Washington University in Saint Louis: Modeling the present and future distribution of whitebark pine. Ph.D. committee member.

Masters' students

Sarah Naughtin, University of British Columbia, M.S. student: Masters' committee member, 2025-present.

Riley Adams, University of Missouri—Saint Louis, M.S. student: Master's committee member, 2025-present.

Post-baccalaureates

Lauren Jenkins, NSF REPS Scholar, Missouri Botanical Garden— *Comparison of methods for estimating past glacial refugia and subsequent migration by trees: Pollen-vegetation models versus ecological niche modeling*. 2021-2022.

Renata Diaz, Missouri Botanical Garden—Testing the ability of threat mapping to predict actual threats to rare species. Summer 2017.

Undergraduate students and interns

Nia Richardson, Howard University—*Effects of global change on stomatal density and size*, Missouri Botanical Garden's NSF-funded Research Experience for Undergraduates: Project co-mentor. 2024 + ongoing mentoring through present.

Léa Vilna-Santos, Washington University in Saint Louis—*Effects of global change on stomatal density and size*. Independent research project: Project co-mentor. 2023.

Evelyn Guerro, University of Texas Permian Basin—*Conservation of urban pollinator networks*, Missouri Botanical Garden's NSF-funded Research Experience for Undergraduates: Project co-mentor. 2022.

Marina Strong, Lake Forest College—*How easy is it to purchase federally-endangered plants from the Internet?* Missouri Botanical Garden's NSF-funded Research Experience for Undergraduates: Project mentor. 2022.

Kelsey Bartlett, George Washington University—*Biogeography of* Asclepias *and phenology of* Leavenworthia: Independent research project: Project co-mentor. 2020-2021.

Tom Collins, University of Missouri—Saint Louis: *Adapting a cellular automaton model of urban growth to reflect adoption of autonomous vehicles*: Independent research project: Project co-mentor. 2018

Renee Klann, Smith College, Missouri Botanical Garden's Research Experience for Undergraduates—*White hat hacking the biodiversity-base: How well could would-be poachers use publicly accessible databases to locate sensitive populations of vulnerable species?* Independent research project: Project mentor. 2018.

High school students

Axel Cortes, Collegiate High School—*Change in phenology and stomatal densities of iconic Midwestern plants resulting from climate change*

Other mentoring

Mentor, International Biogeography Society Mentorship Virtual Meeting, May 2021.

Continuing Professional Education

NSF Convergence Research (CORE) Fellow, 2023: "Tackling Climate-Induced Challenges with Artificial Intelligence"

"Computing in the Cloud" Tutorial at National Institute for Mathematical and Biological Synthesis (NIMBioS), Knoxville, Tennessee.

"Visual Presentation of Data and Information," Workshop by Prof. Edward Tufte, Harvard.

"Bayesian Modeling, Inference, and Prediction," Workshop by Prof. David Draper, UC Santa Cruz.

Santa Fe Institute's Complex Systems Summer School, Qingdao University, Qingdao, Shandong, China.

Project WET Certification: Aquatic resource education workshop for teachers.

Project WILD Certification: Environmental education workshop for teachers.

Service to the Scientific Community: Reviewing

Grant proposals:

National Science Foundation, Panelist Reviewer (2022).

National Science Foundation, Ad Hoc Reviewer (2022, 2024).

Institute for Museum and Library Services, Field Reviewer (2016, 2017, and 2018).

US Geological Survey, White Paper Reviewer (2013).

California Energy Commission, White Paper Reviewer (2012).

National Science Foundation, Ad Hoc Reviewer (2008).

Serrapilheira, Ad Hoc Reviewer (2017).

Conference abstracts:

North American Congress of the Society for Conservation Biology (2012).

International Biogeography Society's Early Career Biogeographers Conference (2020).

Career advancement and professional review:

Ohio University (2025)

University of California, Davis (2018)

Humboldt State University (2018)

US Geological Survey (2021)

Conservation reports:

External reviewer for US Fish & Wildlife Service's "Species Status Assessment of *Leavenworthia crassa*," a federally endangered species (2019).

External reviewer for US Geological Survey's "Effects of Climate Change on Midwestern Ecosystems: Central Interior Oak Forest, Woodland and Savanna" (2026).

Peer-reviewing of scholarly manuscripts:

Associate Editor of *Biological Invasions* (2016-2020). Managing editor for 28 manuscripts.

Reviewing for scientific journals (number of reviews, re-reviews not counted, in order of total number of manuscripts):

Diversity & Distributions (11)
Ecography (11)
Ecology Letters (6)
Journal of Biogeography (5)
Methods in Ecology & Evolution (4)
Ecological Modeling (3)
Ecology (3)
Ecosphere (3)
Global Change Biology (3)
Global Ecology & Biogeography (3)
Nature Ecology & Evolution (3)
American Journal of Botany (2)
Ecological Informatics (2)
Ecology & Evolution (2)
Frontiers in Biogeography (2)
PLoS ONE (2)
Proceedings of the National Academy of Science USA (2)
Annales Zoologici Fennici (1)
Applications in Plant Sciences (1)
Basic & Applied Ecology (1)
Biodiversity Informatics (1)
Biological Invasions (1)
Biological Conservation (1)
Biological Journal of the Linnaean Society (1)
Biological Reviews (1)
Diversity (1)
Natural Areas Journal (2)
Natureza & Conservação (2)
New Phytologist (2)
Oecologica (2)
Oikos (3)
Ecosphere (1)
Landscape Ecology (1)
Ornithology (1)
Plant Biology (1)
Plant Ecology (1)
Plant Ecology & Diversity (1)
Plants, People, Planet (1)
Proceedings of the Royal Society of London B (1)

Annals of the Missouri Botanical Garden (2)
Biodiversity & Conservation (2)
Community Ecology (2)
Climatic Change (1)
Conservation Biology (1)
Conservation Science & Practice (1)
Current Zoology (1)
Science (1)
Scientific Reports (1)
Tropical Conservation Science (1)

Public Service, Outreach, and Press

Public Service

Board member of Trees for Life, International, an international development non-profit organization that empowers people to educate, feed, and nurture themselves. Since 2011.

Leadership Team of BiodiverseCity Saint Louis, a network coordinating environmentally-oriented organizations in the Greater Saint Louis Region. 2020-2023.

Press

Saint Louis Public Radio: St. Louis just missed breaking the all-time heat record in 2024, 2025-01-13.

Saint Louis's FirstAlert4 TV channel: Global experts declare 2024 the warmest year on record, St. Louis could continue to feel the heat, 2025-01-10.

St. Louis Magazine: Three MOBOT scientists on how warming could affect St. Louis' flora, 2022-10-24.

FOX 2 TV: Interview on project for scaling between micro- and macroclimate, 2020-10-15.

St. Louis Public Radio's live *On the Air*: Botanical Garden Looks at Effects of Climate Change on Midwest Plants, 2019-09-27.

Smithsonian Magazine: To Save Endangered Species, Should We Bring Them into Our Cities? (Rachel Kaufman, 2018-10-23).

Outreach

"Pen-and-paper niche modeling to teach about the effects of climate change on trees" lessons and lesson development for high school science teachers, Missouri Botanical Garden, 2022.

"Climate change and biodiversity: Up or out" talk for Missouri Master Naturalists, St. Charles, St. Louis and Franklin Counties Chapter, 2016.

"Climate and Change" talk to Waterloo Rotary, Waterloo, Illinois, 2014.

"Climate Change and Biodiversity" talk to docents at the St. Louis Zoo, St. Louis, Missouri, January 2014.

"Rowing with the Tide: Species' Responses to Climate Change" talk for the Webster Groves Nature Study Society, Webster Groves, Missouri, December 2012.

"Climate Change, Climate Science, and Biodiversity" talk for EcoAct high school students, Saint Louis, Missouri, 2012, 2013, and 2014.

"Mapping Biodiversity: From the Species at Your Feet to the Biodiversity of the World" Public outreach talk at University of California's McLaughlin Natural Reserve, Lower Lake, California, April 2007.

"Fractals: The Shape of Life" display for the Lake County Fair, Lake County, California, August 2007.

"Nature Notes" KBUT weekly summer radio program on ecology and geology co-hosted with Mel Harte. Crested Butte, Colorado, Summer 2005.